

Computer Model Test – 10

Computer Engineering Full Test Computer Model Test – 10

66 of 80 Questions answered correctly

1. In a series circuit with a fixed voltage, if the resistance is increased, how is the power dissipated across the load affected?

- It increases proportionally
- It decreases proportionally
- It remains constant
- It becomes zero

2. Which of the following statements is true regarding Kirchhoff's Voltage Law (KVL)?

- The sum of the currents entering a junction is equal to the sum of the currents leaving the junction.
- The sum of the voltage drops in a closed circuit is equal to the sum of the voltage rises.
- The total voltage in a series circuit is equal to the sum of the individual voltages across each component.
- Both b and c are true.

3. In an RLC circuit, what is the condition for resonance to occur?

- The inductive reactance must equal the capacitive reactance.
- The total impedance must be zero.
- The resistance must be maximum.
- The power factor must be equal to 1.

4. What is the main advantage of using Thevenin's theorem in circuit analysis?

- It simplifies the calculation of power dissipation.
- It allows for the calculation of complex waveforms.
- It reduces the circuit to a simple equivalent circuit for easier analysis.
- It eliminates the need for passive components.

5. In a circuit with a capacitor and resistor in series, if the frequency of the AC supply increases, what happens to the capacitive reactance (X_c)?

- It increases
- It decreases
- It remains constant
- It becomes zero

6. Which of the following configurations of a BJT is used for high-frequency applications due to its lower capacitance?

- Common Base
- Common Emitter
- Common Collector
- Emitter Follower

7. What is the binary equivalent of the decimal number 13?

- 1010
- 1110
- 1101
- 1011

8. How many select lines are required for an 8-to-1 multiplexer?

- 2
- 3
- 4
- 8

9. The main difference between combinational and sequential logic circuits is that sequential circuits:

- Require a clock signal
- Have no feedback
- Depend only on current inputs
- Are faster than combinational circuits

10. Which of the following is true about microprocessors?

- It has an internal memory
- It has interfacing circuits
- It contains ALU, CU, and registers
- All of the above

11. The continuous transfer may be interrupted by an external device by pulling down the signal

- HRQ
- DACK (active low)
- DACK (active high)
- HLDA

12. What is the purpose of an interrupt in a microprocessor system?

- To initiate data transfer
- To stop all processes
- To execute a specific subroutine
- To increase processing speed

13. Which of the following is a fundamental data type in C programming?

- String
- Array
- Integer
- Class

14. In C programming, what does a pointer store?

Integer value
Memory address
String value
Function definition

15. Which access specifier in C++ allows access to all member functions?

Public
Private
Protected
Friend

16. Which feature of object-oriented programming allows a function to be used in more than one way?

Encapsulation
Inheritance
Operator overloading
Class

17. Which keyword is used in C++ to define a virtual function?

Abstract
Public
Static
Virtual

18. Which feature of C++ allows a template to be used for generic programming?

Inheritance
Friend function
Class template
Inline function

19. Which unit controls the sequencing of operations in a CPU?

ALU
Control Unit
Memory Unit
Input Unit

20. Which of the following is an internal memory?

Hard Disk
Cache Memory
CD-ROM
USB Drive

21. Which of the following uses Direct Memory Access (DMA)?

CPU only
Peripheral devices
I/O Devices
Multiprocessor

22. What is an embedded system?

A general-purpose computer
A system designed for specific control functions
A high-performance server
A gaming console

23. In a real-time system, which scheduling algorithm is commonly used?

First-Come, First-Serve
Round Robin
Priority Scheduling
Rate Monotonic Scheduling

24. What does the std_logic type represent in VHDL?

A binary value (0 or 1)
A signal that can take on multiple values (U, X, 0, 1, Z, W, L, H)
A floating-point number
A fixed-point number

25. Which OSI layer is responsible for data encryption?

Network Layer
Data Link Layer
Physical Layer
Presentation Layer

26. Which multiple access protocol is used in Ethernet?

CSMA/CD
Token Ring
ALOHA
FDMA

27. Which protocol is used for dynamic routing?

RIP

FTP
SMTP
HTTP

28. Which port number is used by the HTTP protocol?

21
80
25
443

29. Which encryption algorithm is used in securing email communication?

AES
RSA
PGP
DES

30. What is the primary purpose of a firewall?

To protect data from unauthorized access
To store data
To speed up network performance
To allocate IP addresses

31. Which of the following is used to recognize patterns in a finite automaton?

Regular expression
Context-free grammar
Pushdown automaton
Turing machine

32. Which of the following is a characteristic of context-free languages?

They can be generated by finite automata
They can be parsed using a pushdown automaton
They can only be expressed using regular expressions
They cannot have nested structures

33. Which machine can simulate any other machine's computation?

DFA
Turing Machine
Pushdown Automaton
Mealy Machine

34. What is the basic unit for graphics in a raster display?

Line
Pixel
Shape
Vertex

35. When performing 2D transformations, which coordinate system is typically used?

Cartesian coordinates
Polar coordinates
Homogeneous coordinates
Spherical coordinates

36. Which projection method preserves parallel lines in 3D rendering?

Perspective Projection
Parallel Projection
Orthogonal Projection
Isometric Projection

37. Which data structure uses the Last In, First Out (LIFO) principle?

Stack
Queue
Linked List
Tree

38. Which sorting algorithm is based on the divide-and-conquer technique?

Selection Sort
Bubble Sort
Insertion Sort
Merge Sort

39. In a relational database, which key uniquely identifies a record in a table?

Foreign Key
Candidate Key
Primary Key
Composite Key

40. Which property of transactions ensures all operations are completed successfully or none at all?

Atomicity
Consistency
Isolation
Durability

41. What is the purpose of a process control block (PCB)?

To store the data of a process
To keep track of process status and control information
To execute system calls
To manage memory allocation

42. Which page replacement algorithm replaces the page that hasn't been used for the longest time?

FIFO
LFU
LRU
Random Replacement

43. Which software development model uses short iterative cycles?

Agile Model
Waterfall Model
V-Model
Spiral Model

44. Which document specifies the functional and non-functional requirements of a system?

Use Case Document
System Requirement Specification (SRS)
Design Document
Test Plan

45. Which design concept emphasizes dividing software into modules?

Modular Decomposition
Waterfall Design
Structural Design
Procedural Design

46. What type of testing is performed by developers during the coding phase?

System Testing
User Acceptance Testing
Integration Testing
Unit Testing

47. Which diagram represents the interactions between objects in an object-oriented system?

Class Diagram
Sequence Diagram
Collaboration Diagram
Use Case Diagram

48. Which OOP principle allows a derived class to inherit properties from a base class?

Encapsulation
Polymorphism
Inheritance
Abstraction

49. What is the main component of an intelligent agent's architecture?

Input and Output Devices
Sensors and Actuators
Algorithm and Data
Decision Table

50. Which of the following search algorithms is considered a depth-first search?

A* Search
Breadth-First Search
Iterative Deepening Search
Greedy Best-First Search

51. Which knowledge representation technique uses a graph of nodes and arcs?

Propositional Logic
Frame
Semantic Net
Production Rules

52. Which machine learning technique involves a teacher providing labeled data to the system?

Unsupervised Learning
Supervised Learning
Reinforcement Learning
Genetic Algorithm

53. Which neural network is known for its ability to minimize error using backpropagation?

Convolutional Neural Network (CNN)
Hopfield Network

Perceptron
Multilayer Perceptron (MLP)

54. In artificial neural networks, what is the role of an activation function?

To initialize the network
To normalize the input data
To introduce non-linearity into the network
To train the network

55. What type of projection shows an object as it would appear in 3D space using three dimensions?

Orthographic Projection
Pictorial View
Perspective Projection
Isometric Projection

56. Which of the following methods is used to calculate the Net Present Value (NPV) of a project?

Discounted Cash Flow
Payback Period
Internal Rate of Return (IRR)
Depreciation

57. What does a PERT chart help project managers to determine?

Critical Path
Budget
Staffing
Risk Management Plan

58. What does the term “scope creep” refer to in project management?

The expansion of project goals without corresponding adjustments to resources or time
The process of defining project goals
The unexpected increase in project budget
The gradual reduction of project scope

59. What is the role of engineers in societal development?

To only focus on technical solutions
To ensure their work aligns with societal needs and values
To minimize costs regardless of social impact
To prioritize profits over public welfare

60. Which document typically outlines the code of conduct for engineers?

Engineering Ethics Handbook
Code of Practice
Engineering Registration Act
National Engineering Policy

61. In a series circuit with a 12V battery connected to a resistor of 4Ω and a capacitor of 2μF, what is the time constant (τ) of the circuit?

0.02 seconds
0.008 seconds
0.04 seconds
0.1 seconds

62. A three-phase system has a line voltage of 400V. What is the phase voltage in a star (Y) connection?

800V
200.5V
397.65V
346.4V

63. What is the two's complement of -5 in an 8-bit signed number system?

11111011
11111010
11111101
10000011

64. If a JK flip-flop has J=1, K=1, and the previous state was 0, what will be the next state?

0
1
It will toggle
Undefined

65. What will be the output of the following C code?

```
#include <stdio.h>
void func(int *a) {
    *a += 5;
}
int main() {
    int x = 10;
    func(&x);
    printf("%d", x);
    return 0;
}
```

10
15
5
Compilation Error

66. What is the output of the following C++ program?

```
#include <iostream>
using namespace std;
```

```
class Base {
public:
    virtual void show() {
        cout << "Base class" << endl;
    }
};
```

```
class Derived : public Base {
public:
    void show() override {
        cout << "Derived class" << endl;
    }
};
```

```
int main() {
    Base *b; // Base class pointer
    Derived d; // Derived class object
    b = &d; // Assign derived class address to base class pointer
    b->show(); // Call the show function
    return 0;
}
```

Base class
Derived class
Compilation Error
Runtime Error

67. What is the size of the address bus in a microprocessor that can address 64KB of memory?

16 bits
18 bits
32 bits
64 bits

68. In an 8-bit microprocessor, how many different values can be represented?

128
255
256
512

69. In a network, if the data rate is 100 Mbps (megabits per second), how long will it take to transfer a file of size 1 GB (gigabyte)?

80 seconds
60 seconds
100 seconds
180 seconds

70. If a router processes a packet in 5 ms and the transmission delay is 10 ms for each packet over a link, what is the total time taken to transmit 20 packets?

100 ms
150 ms
200 ms
250 ms

71. Which of the following is true regarding the pumping lemma for context-free languages?

It is a property used to prove that certain languages are not context-free
It applies only to regular languages
It guarantees that all context-free languages are regular
It is only applicable to finite automata

72. A rectangle in a 2D space has vertices at (1, 2), (1, 4), (3, 4), and (3, 2). If the rectangle is scaled by a factor of 2 about the origin, what will be the coordinates of the vertex originally at (1, 2)?

(0.5, 1)
(1, 2)
(2, 4)
(2, 1)

73. What is the worst-case time complexity of accessing an element in a hash table?

O(n)
O(1)
O(log n)
O(n log n)

74. In a binary tree with 7 nodes, what is the maximum number of leaf nodes?

3

7
5
4

75.What does the acronym “CRUD” stand for in database management?

Create, Read, Update, Delete
Create, Review, Upload, Download
Connect, Retrieve, Update, Delete
Create, Run, Update, Destroy

76.In a software project, if the estimated number of lines of code (LOC) is 10,000 and the average productivity rate is 50 LOC per hour, how many hours will it take to complete the coding phase?

150 hours
200 hours
250 hours
300 hours

77. In reinforcement learning, what is the purpose of the exploration-exploitation trade-off?

To ensure the agent always chooses the action with the highest immediate reward.
To balance between exploring new actions and exploiting known rewarding actions.
To minimize the total number of actions taken during training.
To ensure that the agent does not learn from its environment.

78. If a neural network has 3 input neurons, 5 hidden neurons, and 2 output neurons, what is the total number of weights in the network?

15
20
24
30

79.If a project has an estimated duration of 12 weeks and requires 3 team members working full-time (40 hours per week), how many total man-hours are allocated for the project?

1,440 hours
1,200 hours
960 hours
480 hours

80.What is the main objective of risk management in project planning?

To avoid all risks at any cost.
To identify, assess, and prioritize risks to minimize their impact.
To eliminate uncertainties in project outcomes.
To ensure that all project tasks are completed on time.